

The risk and return of arbitrage in dual-listed companies: replication and a test under realistic execution

de Jong, Rosenthal & van Dijk (2009), Review of Finance 13: 495-520 · final report, 2026-10-10

Benchmark return, % per month 1.158 ● paper 1.180	Fama-French alpha, % per month 0.794 ● paper 0.718, both at 1%	Return lost trading one close later 56% ● 85% for the Australian twins	Sharpe ratio under our standard 0.16 ● paper's conventions 0.38
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The paper's claims, checked

C1 ✓ Reproduced Large, persistent deviations from parity in every DLC Table II: 58/72 statistics within rounding; deviation series equal to the authors' column to 4.4e-16	C2 ✓ Reproduced Relative returns co-move with local indices and exchange rates Table III: 77/84 within rounding, 48/48 significance marks
C3 ! Partly Threshold arbitrage earns abnormal returns of up to ~10% p.a. after costs reproduced under the paper's conventions; not robust to execution FF3 alphas within 0.15% p.m. (benchmark 0.794 vs 0.718); trading one close later removes 56% of the return	C4 ✓ Reproduced Arbitrage risk is idiosyncratic and fat-tailed, ~30% volatility R² ~0.003; sigma 26-36 vs 31-35; kurtosis high (driven by the Dexia spike)

C5

✗ Challenged

Time zones and currencies do not explain the mispricing (ADR test)

challenged (no ADR test here)

Australian twins (10 h) lose 85% of their return when trades follow the signal by one close: much of the return is non-synchronous noise

C6

– Not tested

Taxes, governance, short-sale constraints explain little

C7

– Not tested

Prices converge within days of a unification announcement

not tested directly; the strategies hold positions into the announcement (D20)

Final report of the quant research lab on paper 2, steps 13-19. Date: 2026-10-10.

- **Paper:** de Jong, Rosenthal & van Dijk (2009), *The Risk and Return of Arbitrage in Dual-Listed Companies*, Review of Finance 13: 495-520.
- **Data:** the authors' own Datastream workbooks for 12 dual-listed companies (DLCs), licensed: aggregates only in this report. Public: FRED 3-month T-bill (DTB3) and Kenneth French's daily factors.
- **Every number below is taken from a committed report or a named MLflow run.**
- The numbers are collected with their sources in `research/reports/dejong_final/report_data.yaml`.
- The HTML and PDF versions of this report are rendered from that file (`quant_lab.reporting.dejong_final`).

Summary

Is the paper reproducible?

Yes. From the authors' own data, the deviations (Table II), the comovement regressions (Table III), the arbitrage strategies (Tables IV-V) and their abnormal returns (Table VI) reproduce closely. The benchmark strategy earns 1.158% per month (paper 1.180) with a Fama-French alpha of 0.794% (paper 0.718), significant at 1%.

Is its result robust?

No. The paper trades at the close on which its signal appears. Trading one close later, as any real desk must, removes 56% of the benchmark return (1.20 → 0.53% per month, Dexia spike cleaned), and 85% for the Australian twins, whose markets close 10 hours apart.

What survives under the lab's standard?	A calendar-time Sharpe ratio of about 0.16 before realistic costs, concentrated in twins that trade in one time zone. On the two pairs with the lab's full cost model, Sharpe ratios of -0.07 to 0.10 after costs, on 3-5 positions per run.
Agreement with paper 1	The same conclusion as the Maymin replication (steps 1-12): the mispricing is real and persistent, and execution and costs absorb the gain from trading it.

1. Question

Why do the prices of twin shares, claims on the same cash flows, deviate from parity for years in large and liquid companies? The paper measures the risk and return of simple arbitrage strategies after realistic costs and margin requirements, and concludes that the risk deters arbitrage.

This lab asked two questions:

1. Can the paper's results be reproduced from the authors' own data?
2. What is left of the arbitrage return under the lab's standard of execution: no look-ahead, trades after the signal, realistic costs?

2. The paper and its claims

research/extraction/dejong_dlc.md

#	Claim	Verdict here
C1	Every DLC shows large, persistent deviations from parity	Reproduced (Table II)
C2	Relative returns co-move with local indices and exchange rates	Reproduced (Table III)
C3	Threshold strategies earn abnormal returns of up to ~10% p.a. after costs and margin requirements	Reproduced under the paper's conventions; not robust to execution (Tables IV-VI, step 18)
C4	The risk is idiosyncratic and fat-tailed, ~30% volatility	Reproduced (Table VI)
C5	Time zones and currencies do not explain the mispricing (an unreported ADR test)	Challenged. Not tested with ADRs here, but the non-synchronous Australian twins lose 85% of their return under next-close execution

#	Claim	Verdict here
C6	Taxes, governance and short-sale constraints explain little	Not tested
C7	Prices converge within days of a unification announcement	Not tested directly. The strategies turn out to hold positions into the announcement (D20)

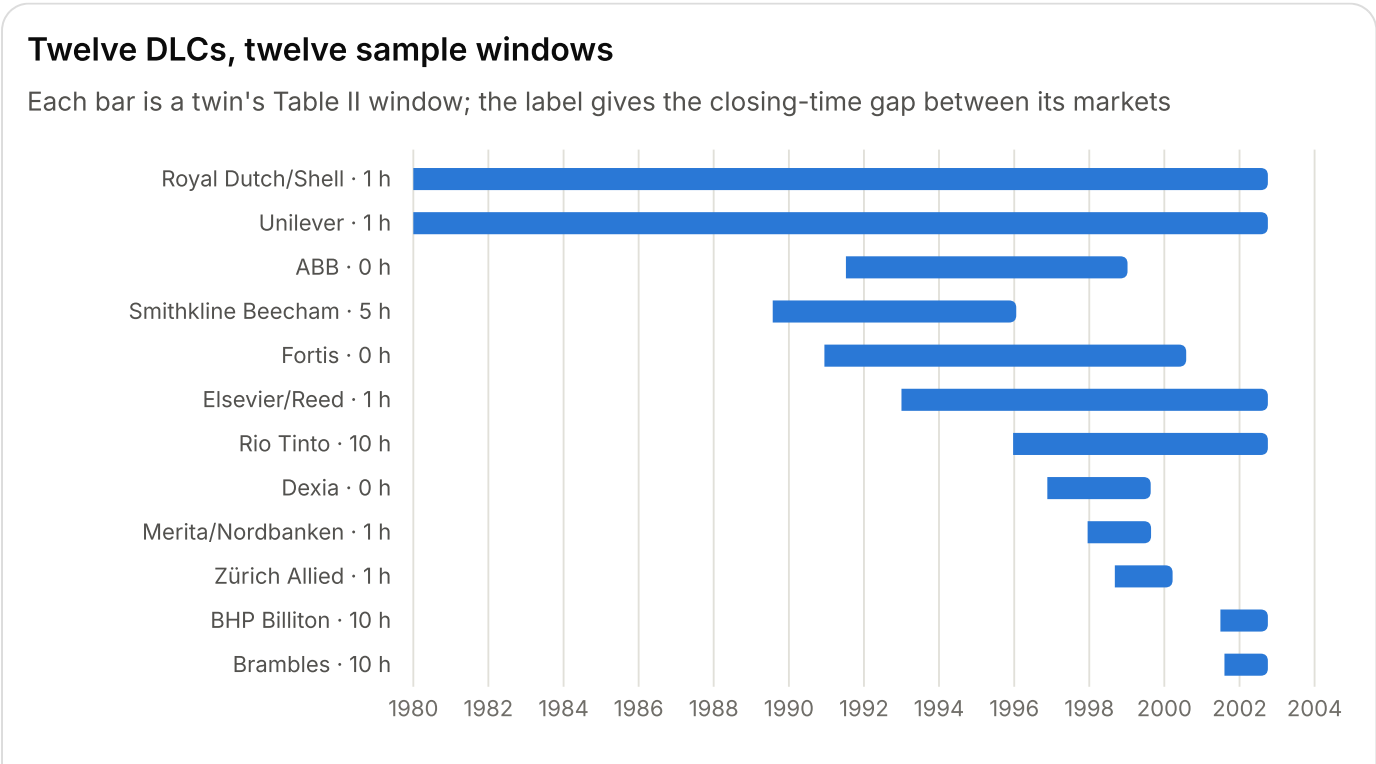
The paper is in-sample by design: thresholds and horizons are chosen and reported on the same 1980-2002 data.

3. Data and sample design

research/reports/data_decisions.md (2026-10-07), data/metadata/

Twins and windows. All 12 DLCs, each over the paper's Table II window: from the merger (1980 for RD/Shell and Unilever) to 20 trading days before the unification announcement, or to 2002-10-03.

Closing-time gap	Twins
0 hours	ABB, Dexia, Fortis
1 hour	Royal Dutch/Shell, Unilever, Elsevier/Reed, Merita/Nordbanken, Zürich Allied
5 hours	Smithkline Beecham
10 hours	Rio Tinto, BHP Billiton, Brambles



Twin	Countries	Gap (h)	Window	Unified in sample
Royal Dutch/Shell	NL/UK	1	1980-01-01..2002-10-03	no
Unilever	NL/UK	1	1980-01-01..2002-10-03	no
ABB	CH/SE	0	1991-07-08..1999-01-07	yes
Smithkline Beecham	UK/US	5	1989-07-26..1996-01-22	yes
Fortis	NL/BE	0	1990-12-12..2000-07-31	yes
Elsevier/Reed	NL/UK	1	1993-01-01..2002-10-03	no
Rio Tinto	AU/UK	10	1995-12-21..2002-10-03	no
Dexia	FR/BE	0	1996-11-19..1999-08-20	yes
Merita/Nordbanken	FI/SE	1	1997-12-15..1999-08-23	yes
Zürich Allied	CH/UK	1	1998-09-07..2000-03-20	yes
BHP Billiton	AU/UK	10	2001-06-29..2002-10-03	no
Brambles	AU/UK	10	2001-08-07..2002-10-03	no

Source: conf/dlc/*.yaml; research/reports/data_decisions.md

The windows reach 2002-10-03. The user approved these windows for this replication. The 2000-02 period was the out-of-sample period of paper 1 (used once, in step 11). So paper 2 has no out-of-sample period in this dataset, and every run that reads 2000-02 rows is tagged

`unlock_oos=true` .

Inputs are checked against the authors' own columns.

- The deviation from parity, $d = \ln(P_A/P_B) - \ln(R)$, rebuilt from prices, FX and the workbook's theoretical ratio, equals the authors' column on every row of all 12 twins. The largest error is $4.4e-16$.
- The regression returns equal our panels' total-return returns: 48,978 checked, largest error $2.2e-15$.

Public data.

- FRED DTB3, md5 `7edbf761...` .
- French daily factors, md5 `95ef09e3...` .

Both were downloaded by the user and versioned with DVC.

4. Methods and rule classifications

research/assumptions/dejong_dlc.md (D1-D25), conf/dlc/ , conf/dejong/default.yaml

The paper fixes its parameters, such as thresholds, costs, margins and rates, but not every mechanic, and the workbooks contain no simulation sheets. Each open convention was identified from the paper's own numbers, one at a time, and tagged:

Area	Convention	How it was settled
Table III	e.r. is B's currency per unit of A's; R^2 unadjusted; leads and lags across the window edge; RD/Shell index without Royal Dutch; Dexia without the spike days; Smithkline on Datastream dates (D13-D17)	each reproduces every value of its table; the literal reading is kept as variant <code>as_stated</code> (59/84)
Table III	Newey-West: EViews default, fixed lag, no prewhitening (D18)	[IMPLEMENTATION-ASSUMPTION], chosen before comparing; 48/48 significance marks
Tables IV-V	entry on a crossing of the buy threshold (D6); exit at the sell threshold (D4); no new position within 22 days, the month the proceeds sit in the T-bill (D5)	position counts and holding periods
Tables IV-V	trades at the close on which the signal appears (D4)	only this timing gives the paper's all-positive returns for the Australian twins; its own one-day-delay case halves alpha
Tables IV-V	P&L in one currency (D19); half spread at exit too (D3); the short-sale margin deposit earns nothing (D21)	per-twin minima and totals
Tables IV-V	unified twins: open positions held until the first trading day after the announcement (D20)	holding periods of ABB, Dexia, Merita, Zürich
Tables IV-V	margin calls per leg (D10)	[IMPLEMENTATION-ASSUMPTION]; not identified : per leg gives 27 positions with calls, pooled 1, the paper 14
Table VI	stacked position-days (D11); every day kept, factors 0 on US holidays (D22); sigma = daily sd $\times 22$, the paper's "% per month" units (D23)	the paper's "# Days" and S&P 500 sigma

Area	Convention	How it was settled
Table VI	IAPM (D25)	not replicable: Datastream's World index is not in the data

About 30 combinations of conventions were examined to identify Tables IV-V. These are replication conventions, not strategy parameters: no threshold, horizon or cost was chosen on the data.

Our standard (step 18) keeps the paper's rule and changes its conventions one at a time:

1. clean the Dexia spike;
2. trade on the close after the signal;
3. no position held into an announcement, positions open at the end marked;
4. no T-bill padding or lockout.

It also runs the rule as lab strategy `dejong_threshold` (`research/specs/replication/dejong_threshold.md`) through the lab's engine, with liquidity-scaled costs.

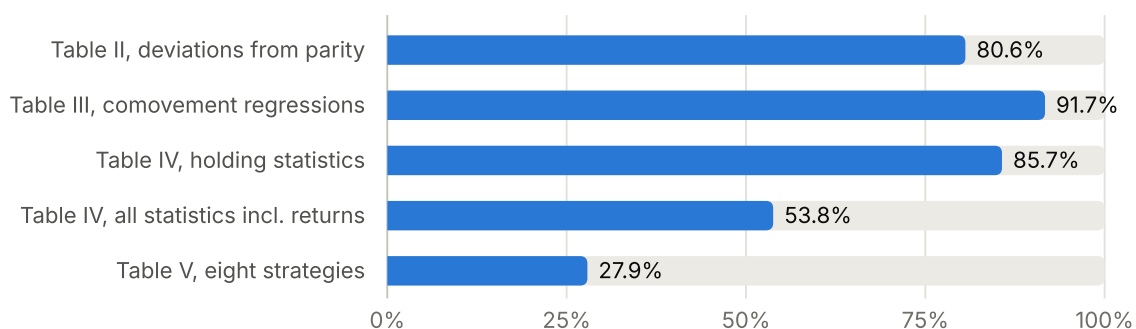
5. Results

5.1 Reproduction of the paper's tables

Table	Within rounding	Notes
II, deviations from parity	58 / 72	Rio Tinto mean sign (- 1.90 vs +1.90, a slip in the paper); ABB st. dev.; Dexia and Fortis differ throughout (data versions)
III, comovement	77 / 84 , 48/48 significance marks	ABB one observation fewer; Brambles df 292 vs 293; Merita -0.317 vs -0.371 (transposed digits); Smithkline DW
IV, holding statistics	78 / 91	position counts exact for 11 of 12 twins (Reed one extra)
IV, all statistics	91 / 169	no single return value within rounding
V, eight strategies	29 / 104	positions and cut-offs within 4 of the paper everywhere
VI, FF3	every alpha within 0.15% p.m.	significance marks agree for 5 of 8 strategies

How much of each table reproduces

Share of the paper's printed statistics our replication matches within rounding

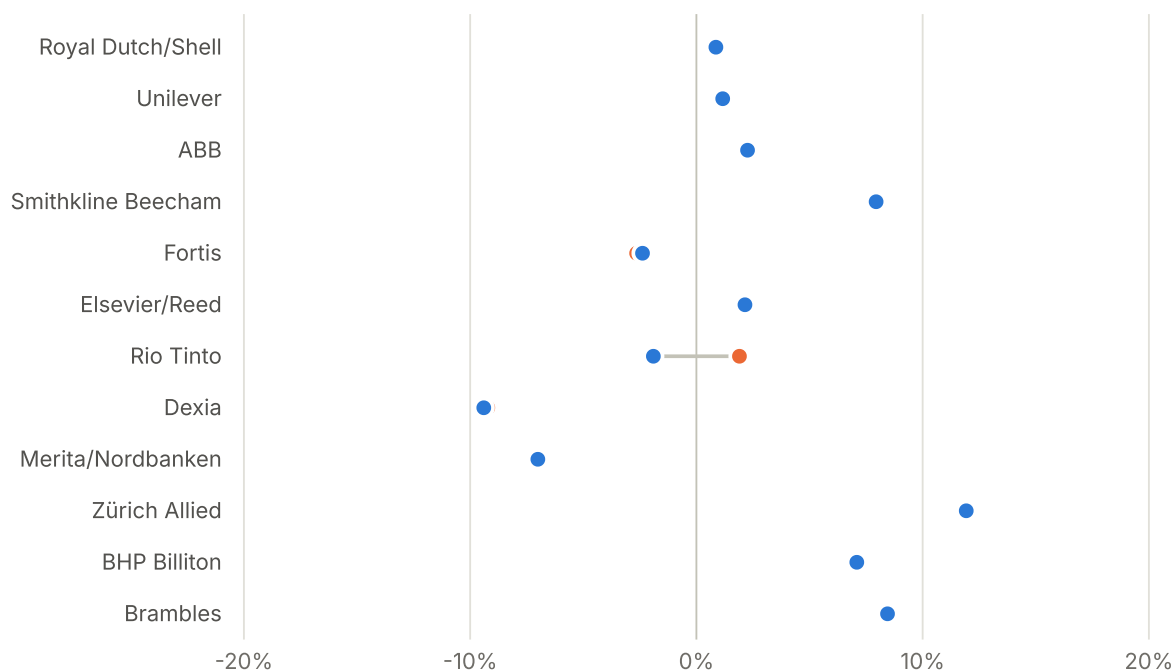


Source: research/reports/dejong_table2.md, dejong_table3.md, dejong_tables45.md, dejong_table6.md

Table II: mean deviation from parity

Mean log deviation in %, ours and the paper's, per twin. Rio Tinto differs only in sign; Dexia and Fortis in data version

● Ours ● Paper



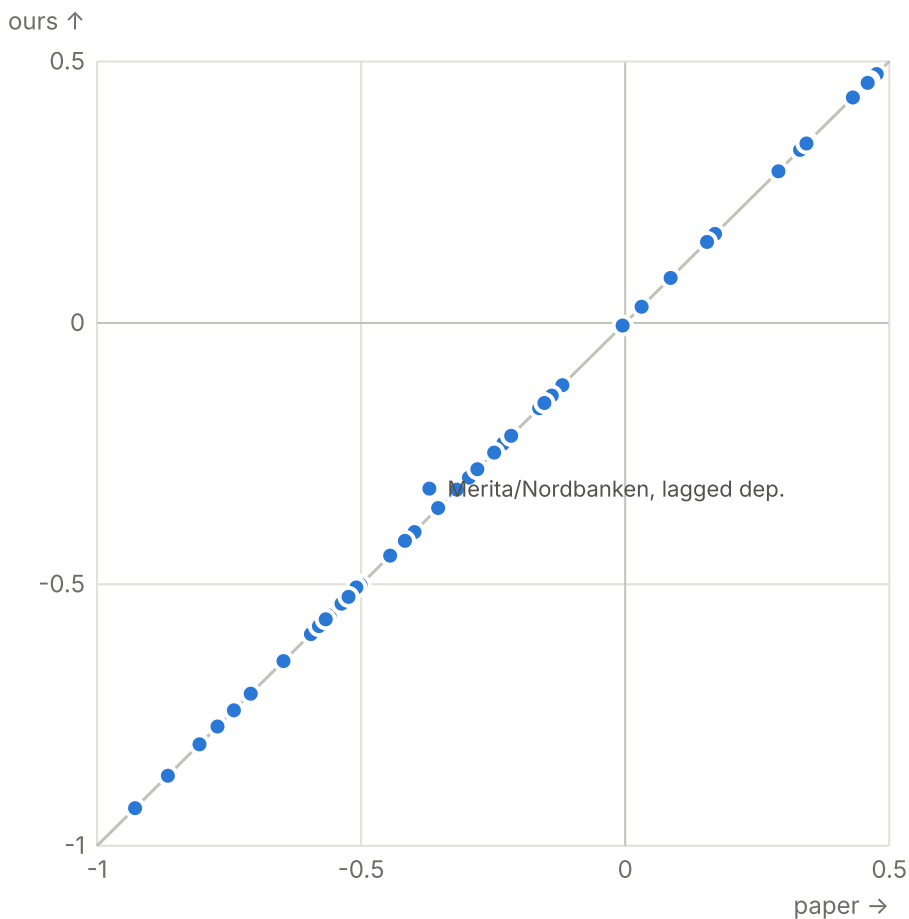
Twin	Mean (ours / paper)	Mean abs.	St. dev.
Royal Dutch/Shell	0.86 / 0.86	10.04 / 10.04	12.71 / 12.71
Unilever	1.16 / 1.16	8.99 / 8.99	11.41 / 11.41
ABB	2.26 / 2.26	8.91 / 8.91	10.02 / 10.17
Smithkline Beecham	7.94 / 7.94	8.10 / 8.10	4.09 / 4.09

Twin	Mean (ours / paper)	Mean abs.	St. dev.
Fortis	-2.38 / -2.64	4.52 / 4.56	5.02 / 4.90
Elsevier/Reed	2.15 / 2.15	8.88 / 8.88	9.20 / 9.20
Rio Tinto	-1.90 / 1.90	4.11 / 4.11	4.76 / 4.76
Dexia	-9.40 / -9.22	9.47 / 9.33	3.56 / 3.67
Merita/Nordbanken	-7.01 / -7.01	7.07 / 7.07	3.19 / 3.19
Zürich Allied	11.93 / 11.93	11.93 / 11.93	3.47 / 3.47
BHP Billiton	7.09 / 7.09	7.09 / 7.09	2.26 / 2.26
Brambles	8.45 / 8.45	11.32 / 11.32	11.32 / 11.32

Source: research/reports/dejong_table2.md

Table III: 48 coefficient sums on the 45-degree line

Each dot is one sum of coefficients (index 1, index 2, exchange rate, lagged dependent) for one twin, ours against the paper's; labelled where they differ



Twin	R ²	Index 1	Index 2	Exchange rate	Lagged dep.
Royal Dutch/Shell	0.242 / 0.242	0.346 / 0.346	-0.501 / -0.501	-0.806 / -0.806	-0.231 / -0.231
Unilever	0.146 / 0.146	0.170 / 0.170	-0.560 / -0.560	-0.595 / -0.595	-0.216 / -0.216
ABB	0.156 / 0.155	0.436 / 0.433	-0.400 / -0.399	-0.506 / -0.509	-0.119 / -0.119
Smithkline Beecham	0.132 / 0.132	0.086 / 0.086	-0.248 / -0.248	0.031 / 0.031	-0.299 / -0.299
Fortis	0.104 / 0.104	0.476 / 0.476	-0.537 / -0.537	-0.580 / -0.580	-0.163 / -0.163
Elsevier/Reed	0.197 / 0.197	0.331 / 0.331	-0.417 / -0.417	-0.772 / -0.772	-0.319 / -0.319
Rio Tinto	0.272 / 0.272	0.431 / 0.431	-0.741 / -0.741	-0.524 / -0.524	-0.296 / -0.296
Dexia	0.100 / 0.100	0.290 / 0.290	-0.324 / -0.324	-0.319 / -0.319	-0.216 / -0.216
Merita/Nordbanken	0.246 / 0.246	0.463 / 0.463	-0.445 / -0.445	-0.139 / -0.139	-0.317 / -0.371
Zürich Allied	0.091 / 0.091	0.155 / 0.155	-0.354 / -0.354	-0.928 / -0.928	-0.153 / -0.153
BHP Billiton	0.397 / 0.397	0.459 / 0.459	-0.709 / -0.709	-0.647 / -0.647	-0.280 / -0.280
Brambles	0.288 / 0.288	0.343 / 0.343	-0.866 / -0.866	-0.567 / -0.567	-0.005 / -0.005

Source: research/reports/dejong_table3.md

Benchmark strategy (10%/5%/12 months), ours / paper:

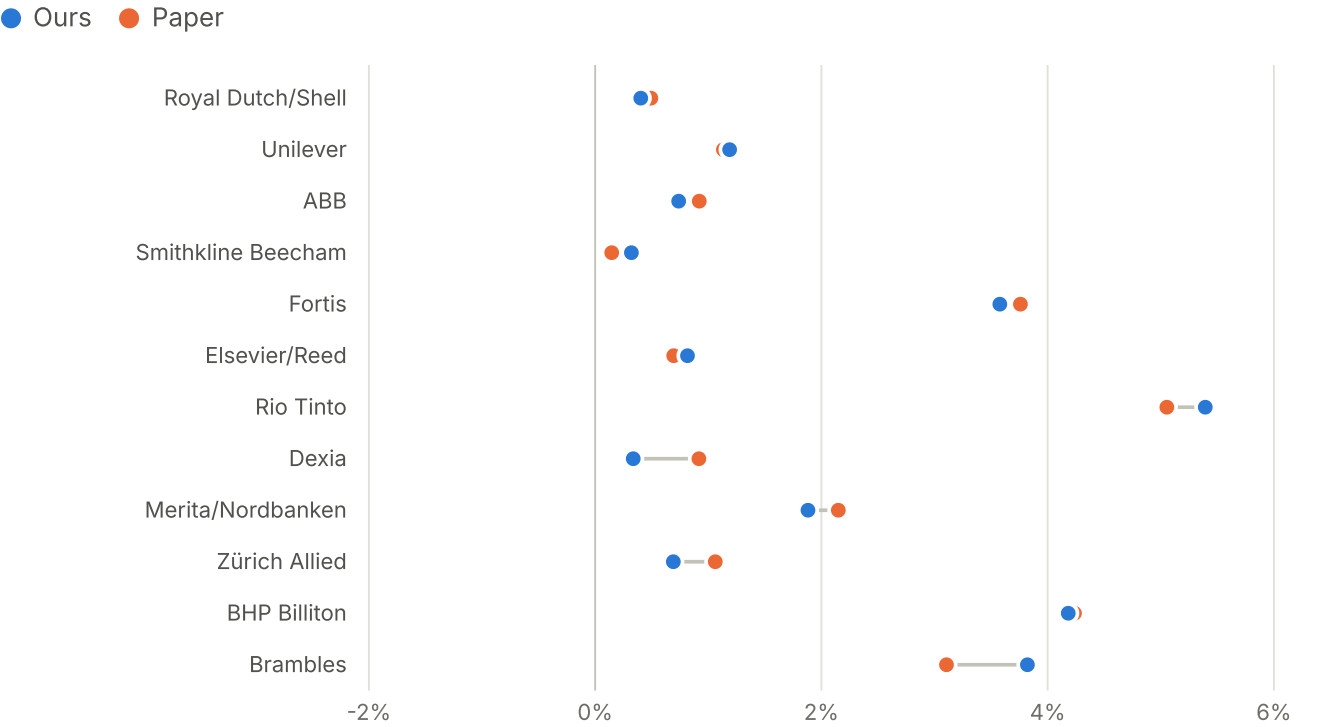
Statistic	Ours	Paper
Positions	128	127

Statistic	Ours	Paper
Weighted mean, % per month	1.158	1.180
Median, % per month	3.697	3.703
FF3 alpha, % per month	0.794 (1%)	0.718 (1%)
Sigma, "% per month" units	29.7	30.7

Sources: `dejong_tables45.md` , run `1d9f0b53` ; `dejong_table6.md` , run `4d73e5e3` .

Table IV: benchmark return per twin

Days-weighted mean return of the 10%/5%/12-month strategy, % per month, under the paper's conventions



Twin	Positions	Mean days	Weighted mean % p.m.	Median % p.m.
Royal Dutch/Shell	15 / 15	126.0 / 126.0	0.403 / 0.492	1.372 / 1.460
Unilever	31 / 31	88.6 / 88.6	1.188 / 1.134	4.497 / 4.346
ABB	10 / 10	96.7 / 96.7	0.739 / 0.920	2.403 / 3.607
Smithkline Beecham	6 / 6	166.8 / 167.7	0.320 / 0.147	0.417 / 0.680
Fortis	18 / 18	32.8 / 32.8	3.578 / 3.760	4.529 / 4.677
Elsevier/Reed	14 / 13	109.4 / 116.2	0.817 / 0.694	1.577 / 1.095

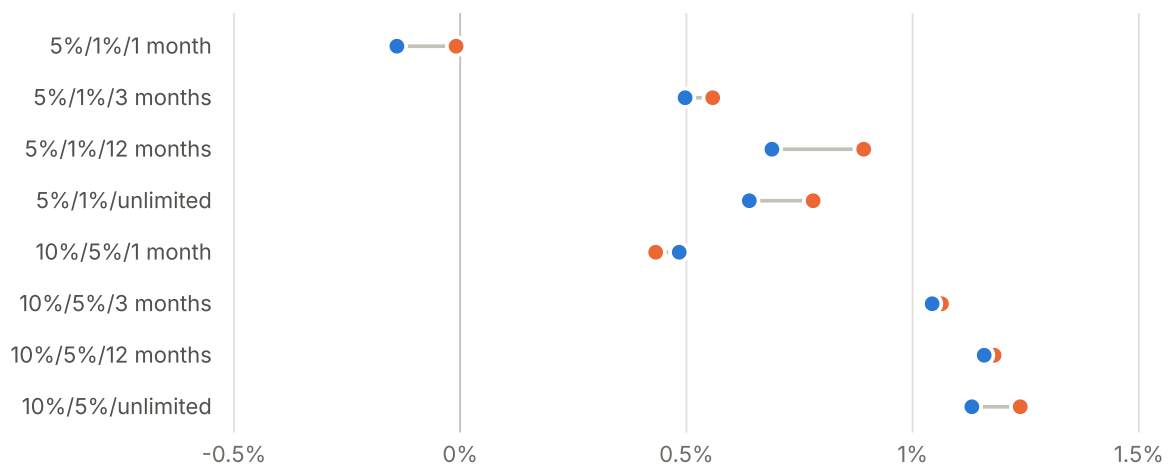
Twin	Positions	Mean days	Weighted mean % p.m.	Median % p.m.
Rio Tinto	13 / 13	24.8 / 24.6	5.394 / 5.054	4.857 / 5.140
Dexia	4 / 4	124.5 / 124.5	0.336 / 0.917	3.296 / 1.179
Merita/Nordbanken	4 / 4	55.3 / 55.3	1.881 / 2.150	4.546 / 2.885
Zürich Allied	3 / 3	133.0 / 133.0	0.691 / 1.062	0.715 / 1.249
BHP Billiton	5 / 5	23.0 / 23.0	4.182 / 4.238	3.719 / 3.703
Brambles	5 / 5	30.6 / 30.6	3.821 / 3.106	4.782 / 3.830

Source: research/reports/dejong_tables45.md (user's run 1d9f0b53bcb749a7a260e1a6b61fdeba)

Table V: eight strategies

Days-weighted mean return, % per month, all twins, buy/sell threshold and maximum horizon

● Ours ● Paper



Strategy	Positions	Weighted mean % p.m.
5%/1%/1 month	313 / 309	-0.140 / -0.009
5%/1%/3 months	203 / 205	0.497 / 0.558
5%/1%/12 months	151 / 156	0.689 / 0.892
5%/1%/unlimited	134 / 138	0.639 / 0.780
10%/5%/1 month	290 / 288	0.484 / 0.432
10%/5%/3 months	185 / 184	1.043 / 1.064
10%/5%/12 months	128 / 127	1.158 / 1.180

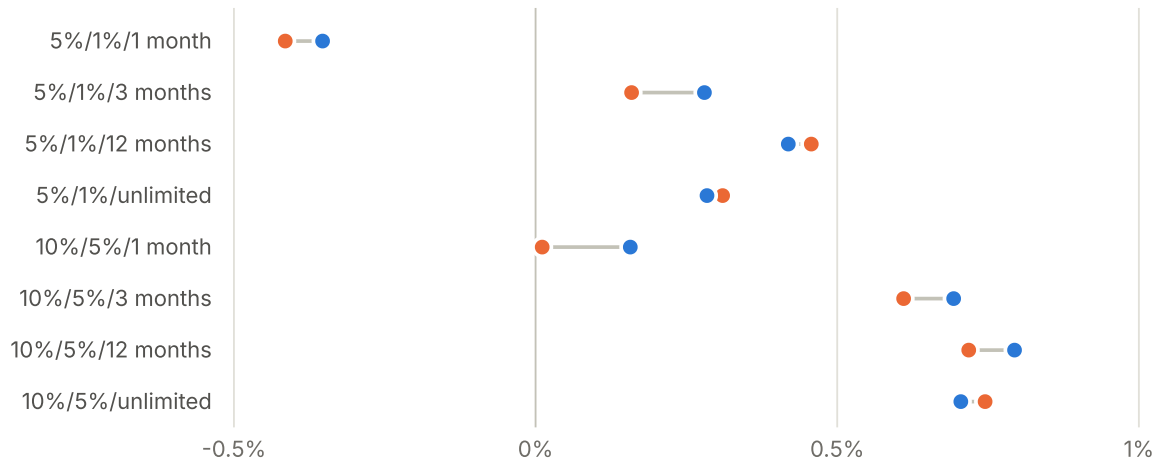
Strategy	Positions	Weighted mean % p.m.
10%/5%/unlimited	113 / 112	1.131 / 1.238

Source: research/reports/dejong_tables45.md (user's run 1d9f0b53bcb749a7a260e1a6b61fdeba)

Table VI: Fama-French alpha

FF3 alpha of the pooled position-days, % per month; significance in the table view (a 1%, b 5%, c 10%)

● Ours ● Paper



Strategy	Alpha % p.m.	Sig.	Position - days	σ	σ S&P 500	Kurtosis	1% VaR
5%/1%/1 month	-0.353 /	- /	6871 /	36.3 /	23.8 /	13.4 /	-4.5 /
	-0.415	-	6798	34.6	23.6	12.4	-4.6
5%/1%/3 months	0.280 /	- /	9136 /	35.1 /	23.9 /	13.2 /	-4.2 /
	0.159	-	9148	34.5	23.8	11.6	-4.5
5%/1%/12 months	0.419 /	- /	12931 /	31.9 /	22.8 /	17.3 /	-3.7 /
	0.457	b	13019	33.5	22.6	11.0	-4.2
5%/1%/unlimited	0.284 /	- /	17409 /	25.5 /	22.2 /	29.3 /	-3.1 /
	0.310	b	17505	31.6	22.1	10.8	-4.0
10%/5%/1 month	0.157 /	- /	6380 /	32.7 /	24.3 /	25.6 /	-3.6 /
	0.011	-	6339	31.0	24.0	8.9	-3.8
10%/5%/3 months	0.693 /	b /	8480 /	31.4 /	23.5 /	13.7 /	-3.6 /
	0.610	a	8462	30.7	23.2	8.3	-3.8
10%/5%/12 months	0.794 /	a /	10437 /	29.7 /	23.2 /	21.9 /	-3.4 /
	0.718	a	10422	30.7	22.9	9.6	-3.7

Strategy	Alpha % p.m.	Sig.	Position- days	σ	σ S&P 500	Kurtosis	1% VaR
10%/5%/unlimited	0.705 /	a /	11199 /	26.5 /	22.6 /	30.8 /	-3.2 /
	0.745	a	11210	30.9	22.2	10.2	-3.8

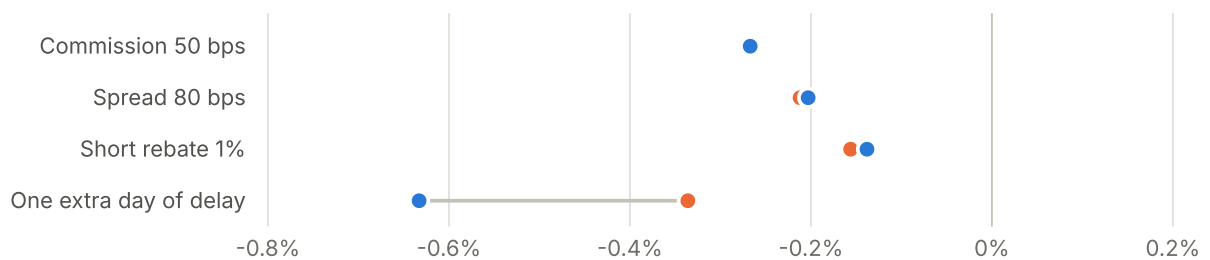
Source: research/reports/dejong_table6.md (user's run 4d73e5e3e6de47a4af3e9be3724ba2a8)

Cost sensitivities reproduce, the delay sensitivity does not. On p. 515 the paper changes one assumption at a time. Our alpha changes by -0.267 (commission 50 bps), -0.203 (spread 80 bps) and -0.138 (rebate 1%), against the paper's -0.265, -0.212 and -0.156. One extra day of delay costs us 0.633 and the paper 0.336.

p. 515 sensitivities: change in the benchmark alpha

Each case minus its own benchmark, % per month. Costs and rebate match; the one-day delay hurts ours twice as much

● Ours ● Paper



Case	Ours	Paper	Ours – benchmark	Paper – benchmark
Benchmark	0.794	0.718	+0.000	+0.000
Commission 50 bps	0.527	0.453	-0.267	-0.265
Spread 80 bps	0.591	0.506	-0.203	-0.212
Short rebate 1%	0.656	0.562	-0.138	-0.156
One extra day of delay	0.161	0.382	-0.633	-0.336

Source: research/reports/dejong_table6.md (user's run 4d73e5e3e6de47a4af3e9be3724ba2a8)

Not reproduced.

- **The margin-call counts:** 27 vs 14 positions.

- **The 5%/1% strategies:** they earn 0.06-0.2% per month less than the paper's. Their best position, 11.7% per month in the paper, does not appear.
- **Brambles, Zürich, Dexia and ABB:** per-twin returns are off by 0.2-0.7% per month.

5.2 The convention waterfall (step 18)

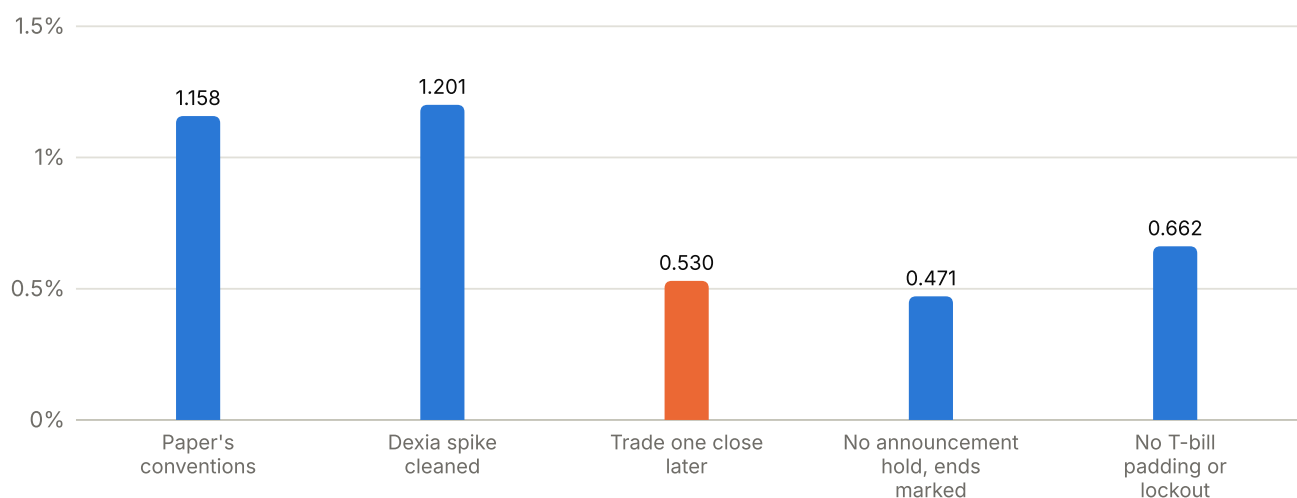
Run `8b36f314`, `dejong_standard.md`. Benchmark 10%/5%/12 months, all twins. Each step includes the ones before it.

Step	% per month (paper metric)	Excess return % p.a. (calendar time)	Sharpe (calendar time)	Positions
Paper's conventions	1.158	5.8	0.38	128
Dexia spike cleaned	1.201	6.0	0.40	128
Trade one close later	0.530	0.5	0.03	128
No announcement hold, ends marked	0.471	0.1	0.01	130
No T-bill padding or lockout	0.662	2.7	0.16	144

From the paper's conventions to ours

Benchmark 10%/5%/12 months, all twins, % per month; each step includes the ones before it. The third step trades one close after the signal

● Cumulative step ● Largest single change



Source: research/reports/dejong_standard.md (user's run 8b36f314afa448319bbe23d897531b56)

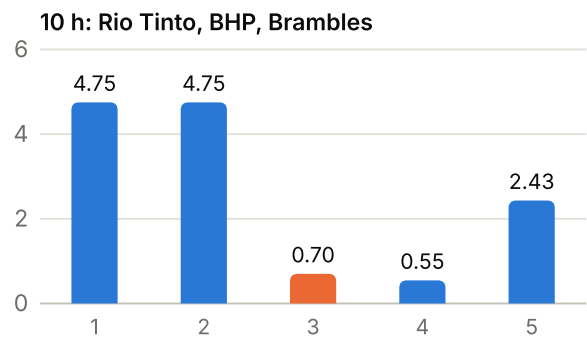
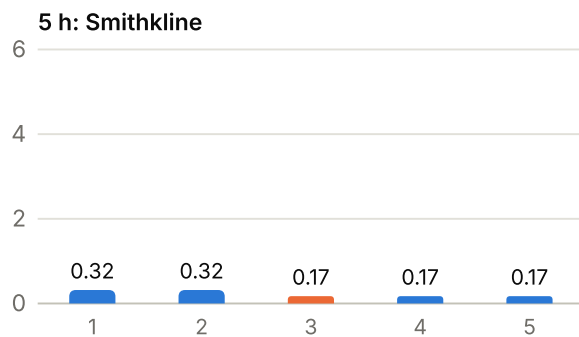
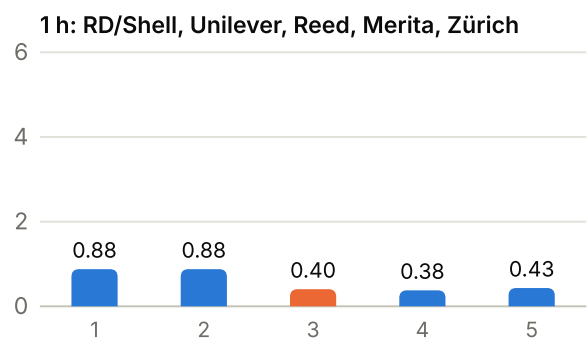
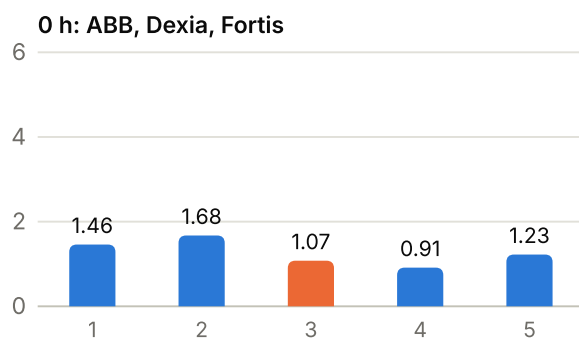
By closing-time gap (% per month, paper's conventions → our standard):

Gap	Paper's conventions	One close later	Our standard
0 hours	1.458	1.075	1.225
1 hour	0.879	0.404	0.433
5 hours	0.320	0.172	0.171
10 hours	4.750	0.698	2.433

Where the return goes: by closing-time gap

Same waterfall per group of twins, % per month, one shared scale. Steps: see the list

● Cumulative step ● Trade one close later



1. Paper's conventions
2. Dexia spike cleaned
3. Trade one close later

4. No announcement hold, ends marked
5. No T-bill padding or lockout

Source: research/reports/dejong_standard.md (user's run 8b36f314afa448319bbe23d897531b56)

Trading one close after the signal is the single largest change in every group. The Australian twins lose 85% of their return at that step. Their deviations, measured on closes 10 hours apart, are partly noise that reverses by the next close. The paper's strategy books that reversal by trading at the same close on which it observes it, which no one can do.

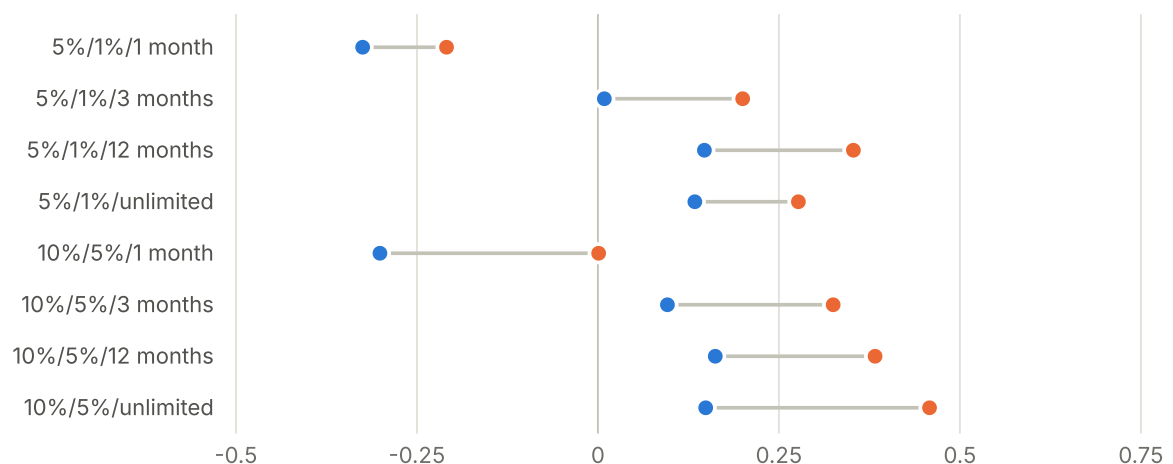
Under our standard, no strategy keeps more than 0.16 of calendar-time Sharpe ratio:

Strategy	Sharpe, paper's conventions → our standard
5%/1%/1 month	-0.21 → -0.33
5%/1%/3 months	0.20 → 0.01
5%/1%/12 months	0.35 → 0.15
5%/1%/unlimited	0.28 → 0.13
10%/5%/1 month	0.00 → -0.30
10%/5%/3 months	0.33 → 0.10
10%/5%/12 months	0.38 → 0.16
10%/5%/unlimited	0.46 → 0.15

Calendar-time Sharpe ratio of all eight strategies

One daily portfolio over all twins, excess over the T-bill: under the paper's conventions and under ours

● Our standard ● Paper's conventions



Source: research/reports/dejong_standard.md (user's run 8b36f314afa448319bbe23d897531b56)

5.3 The lab's engine with liquidity costs

Strategy `dejong_threshold` : the user's eight runs, git clean.

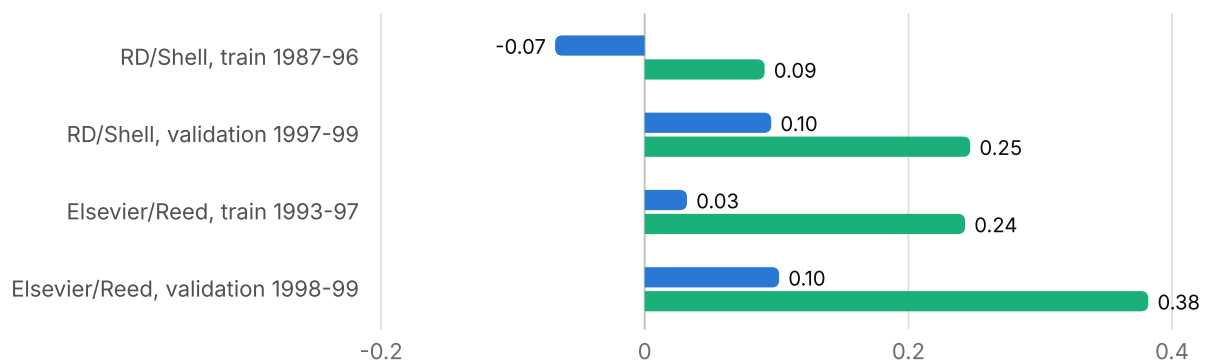
Pair, period	Positions	Sharpe after costs	Sharpe without costs	Median order cost
RD/Shell, train 1987-96	5	-0.07	0.09	22 bps
RD/Shell, validation 1997-99	3	0.10	0.25	29 bps

Pair, period	Positions	Sharpe after costs	Sharpe without costs	Median order cost
Elsevier/Reed, train 1993 - 97	3	0.03	0.24	41 bps
Elsevier/Reed, validation 1998 - 99	5	0.10	0.38	51 bps

The lab's engine: Sharpe ratio after and without costs

Strategy dejong_threshold on the lab pairs' development periods, liquidity-scaled costs; 3-5 positions per run

● After costs ● Without costs



Source: research/reports/dejong_standard.md; MLflow experiments rd_shell.dejong_threshold, reed_elsevier.dejong_threshold

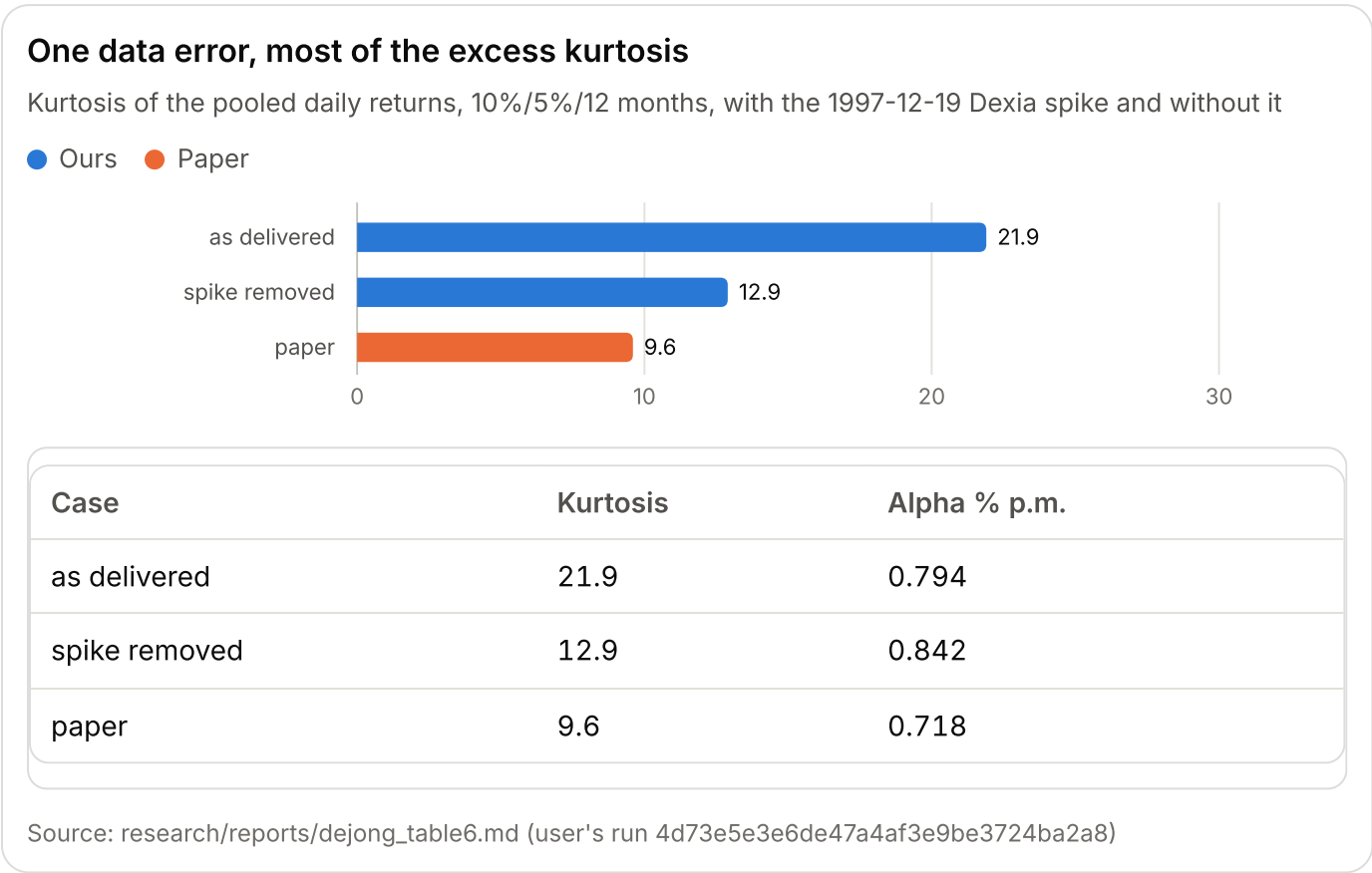
- **Costs are not the main difference.** The lab's costs per order are about the paper's 45 bps per leg; the main difference is timing.
- **After timing, costs remove most of what is left.**
- **No figure here is distinguishable from zero.**

6. Robustness and multiple testing

- **Variants of the paper engine**, each run on all eight strategies (`dejong_tables45.md`):
- Dexia spike removed;
- pooled margin rule;
- price-only P&L;
- one-day delay.

The benchmark weighted mean moves between 1.135 and 1.271 under the first three. Under the one-day delay it falls to 0.489. - **The Dexia spike:** - It decides one position (-1.73% vs -0.01% per

month). - It drives most of the excess kurtosis in Table VI (21.9 → 12.9; paper 9.6). - Together with Table III, this is strong evidence that the authors cleaned it.



- **Multiple testing:**
- The identification examined about 30 convention combinations against the paper's own numbers. Nothing was tuned for performance.
- The step-18 lab runs add 4 trials per pair to the MLflow trial count of `rd_shell` and `reed_elsevier`.
- No deflated Sharpe ratio was computed for paper 2, because no strategy is proposed: the result is a decomposition, not a candidate for trading.

7. Audits

No `/audit-backtest` was run on the paper-2 engine. It reproduces the paper's conventions, including its same-close execution, which an audit would flag as look-ahead in a strategy of ours. The waterfall measures exactly that.

The lab's own engine and `dejong_threshold` inherit the paper-1 audits (`research/reports/audits/2026-10-07-freeze.md`).

The new strategy passes the structural look-ahead test (`tests/structural/test_strategy_causality.py`).

8. Pre-registration and out-of-sample result

Not applicable. Paper 2 is a replication of a published in-sample result on the authors' own data. The sample design decision (§3) states that no out-of-sample period is left in this dataset for paper 2. Nothing in this report is a pre-registered test.

A real test of the step-18 strategy needs data after 2002-10-03.

9. Errata and failed expectations

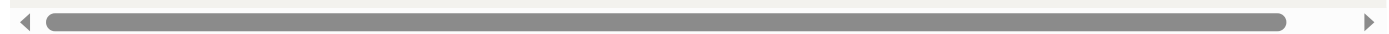
- **Corrected reading of the paper.** Our extraction (step 13) said the strategy samples exclude the unification announcement jumps. Step 16 showed the opposite (D20): open positions in unified twins are held until the first trading day after the announcement, and they collect the jump.
 - **The review's suspicion was confirmed, not refuted.** Before any run, the review ([research/methodology/dejong_dlc.md](#) §2) flagged non-synchronous closes and same-day reversal as a possible source of the return. Step 18 confirms it.
 - **Unresolved:**
 - the paper's margin-call rule;
 - its median convention (inconsistent between twins);
 - its ABB, Dexia and Fortis data versions;
 - the 5%/1% maximum return.
-

10. Limitations

- **In-sample only.** The whole sample is the paper's own period. No out-of-sample claim is possible on this data.
- **Identified conventions are inferences.** They are inferred from matching numbers, not documented by the authors. Where two readings fit, both are reported.
- **The IAPM is not replicated.** The World index is missing; the paper's IAPM alphas lie within 0.05 of its FF3 alphas.
- **The lab's cost model covers two pairs only.** The other ten twins have no validated lab panels (volume and quotes). Their "our standard" figures keep the paper's flat costs.
- **Power.** The lab-engine runs have 3-5 positions each. The pooled waterfall is the informative result.
- **Time-zone groups are small.** 3, 5, 1 and 3 twins; the 5-hour group is a single twin.

11. Reproducibility

```
git clone https://github.com/tsNoordam/quant-lab.git && cd quant-lab
uv sync --locked
uv run dvc pull && uv run dvc repro dlc_ingest # licensed data: needs access to the DVC remote
uv run pytest # real-data pins: tests/integration/test_dejong_*
uv run python -m quant_lab.models.dejong # Tables II-III
uv run python -m quant_lab.backtest.dejong # Tables IV-V
uv run python -m quant_lab.models.dejong_risk # Table VI
uv run python -m quant_lab.models.dejong_standard # step 18 waterfall
uv run python -m quant_lab.reporting.dejong_final # this report as HTML and PDF
```



Pins. Paper-2 headline numbers are pinned in `tests/integration/test_dejong_real.py` and `test_dejong_engine_real.py` :

- Table II/III differences;
- holding statistics;
- the eight weighted means;
- the eight alphas;
- the waterfall.

Paper 1 versus paper 2. `uv run python -m quant_lab.reproduce` covers paper 1's headline numbers only.

There are no freeze tags for paper 2: nothing was pre-registered.

12. Conclusion

The paper is reproducible from its authors' data to a degree that is rare. Most entries of Tables II and III, and most holding statistics of Table IV, match to the printed decimals. The strategy returns and alphas come within about 0.15% per month without matching exactly. Where they differ, the cause is identified or narrowed to a data version or an unidentified convention.

The result it reports, about 9% a year of abnormal return from arbitraging twin shares, depends on trading at the closing price on which the opportunity is observed. One close later, and with the Dexia data error cleaned, about half of it disappears. For the twins whose markets close hours apart, most of it disappears.

What remains is a calendar-time Sharpe ratio of about 0.16 before realistic costs, and around zero after them on the pairs where those costs can be measured.

The mispricing is real and persistent, as both papers say. Like paper 1, this lab finds no tradable edge in it.

Rendered from research/reports/dejong_final_report.md and research/reports/dejong_final/report_data.yaml by quant_lab.reporting.dejong_final. Aggregates only; the underlying Datastream data is licensed.